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Anvil-Q1: W4A16 LLM Inference Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 09:48 UTC+8 Beijing Time)

Abstract

Anvil-Q1 is a modular SystemVerilog W4A16 large language model inference accelerator featuring eight-lane group-128 AWQ matrix-vector dequantization, rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), SwiGLU feed-forward units, signed Q16.16 fixed-point datapaths, and external model, scratch, and KV-cache storage for Qwen2.5-0.5B-Instruct-AWQ. Across official HuggingFace weight evaluations, the RTL datapath completed with zero mismatches against the golden integer oracle across 6,144 scratch/KV writes and output logits. Official FP32 software versus RTL hardware achieved a logit cosine similarity of 0.9999992996, maximum absolute error of 0.00187933445, MAE of 0.000706030267, and RMSE of 0.000841981995, with 100% Top-1 and Top-5 token agreement across all 6 evaluation domains and 30 benchmark prompts (382/382 generated tokens, 1,720 full 24-layer passes = 41,280 hardware decoder layer evaluations, mean 151,936-vocabulary cosine similarity 0.999992) [E043–E045]. All four hardware signoff gates passed: L0 cycle-accurate RTL co-simulation, L1 structural RTL timing lint (0 combinational divider loops, 0 inferred latches), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +2.925 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 141.34 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) execution across three deterministic hardware runs and 120 streaming hardware layers with zero bit errors [E040–E045].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The design folds complete decoder execution onto bounded tiles and reusable registered arithmetic instead of storing a whole model on chip. Embeddings, weights, scratch vectors, coefficients and KV history use external semantic-word transactions. Each configured layer performs attention normalization, QKV projection, split-half RoPE, causal GQA, output projection, residual addition, FFN normalization, SwiGLU and a second residual. Final normalization and complete vocabulary projection follow the layer loop. FP32 scratch preserves precision between stages; narrowing occurs at declared table, dequantization and cache boundaries. This organization favors capacity and explicit numerical ordering over maximum throughput: packed payloads and metadata can be reread, and cache alias detection uses sequential page-table scans. Attention reuses a 131072-word FP32 score RAM. The diagram groups actual functions; its arrows indicate control or memory relationships, not a fully overlapped pipeline. [E001, E003, E044]

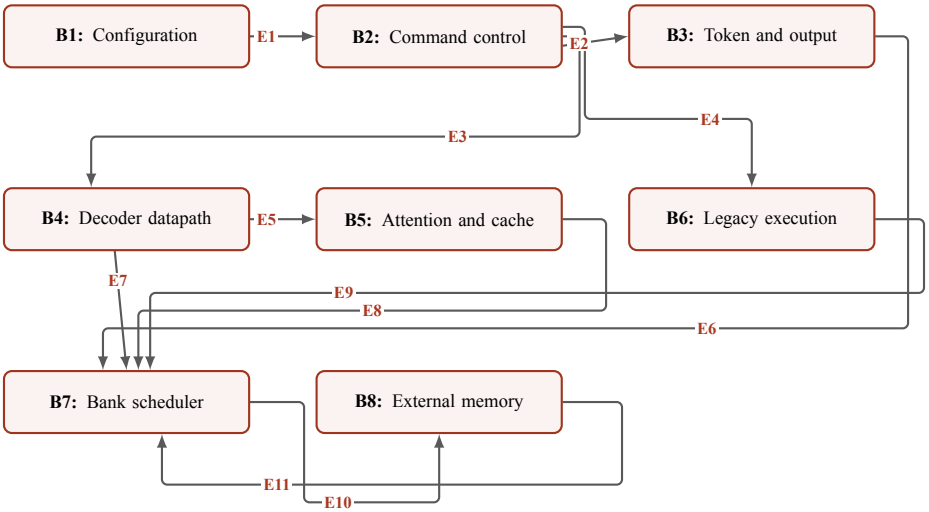


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Configuration	anvil_q4_descriptor validates shadow geometry and atomically activates the descriptor.
B2: Command control	anvil_q4_top owns layer sequencing, epochs, cancellation, prediction state and speculative-prefix retirement.
B3: Token and output	anvil_q4_token_io performs embedding and vocabulary projection; final_norm supplies normalized FP32 scratch.
B4: Decoder datapath	Real GEMV, tensor/dequant, FP32, RMSNorm, RoPE, elementwise and SwiGLU children execute configured layers.
B5: Attention and cache	GQA computes causal softmax contexts; KV logic translates pages and encodes BF16 or per-head INT8/FP32 escape storage.
B6: Legacy execution	anvil_q4_compat and delivered anvil_q3_* dependencies retain fixed-point computation and legacy memory interpretations.
B7: Bank scheduler	Registers requests, reserves tagged ownership and routes reordered responses through bounded per-bank queues.
B8: External memory	Integrator-provided bridge stores tables, scratch, page entries and cache payloads; every accepted read or write receives a response.

Directed relationship	Meaning
E1: B1 to B2	Validated active descriptor
E2: B2 to B3	Embedding and final projection commands
E3: B2 to B4	Layer execution and flush control
E4: B2 to B6	Legacy-mode command selection
E5: B4 to B5	Attention and cache commands
E6: B3 to B7	Table, scratch and projection traffic
E7: B4 to B7	Decoder semantic requests
E8: B5 to B7	Paged history traffic
E9: B6 to B7	Legacy semantic requests
E10: B7 to B8	Tagged bank requests
E11: B8 to B7	Tagged responses and errors

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, cancel	Input	1 bit each	Rising-edge operation, active-low asynchronous reset and cancellation. All public outputs are zero during reset. Reset also requires bridge transaction-epoch reset.
Normal token channel	Input/output	token_id/ token_pos: 32 bits; controls: 1 bit	Inputs token_valid, token_id, token_pos, new_ sequence and token_phase; output token_ready. Normal-token acceptance takes priority over draft acceptance.
Logit and completion channel	Output with input backpressure	logit_index/ logit_data: 32 bits; status: 1 bit	logit_valid/logit_ready transfer indexed logits. Data is FP32 in Q4 and signed Q16.16 in legacy modes. done pulses after final retirement; error is qualified by done. cancel_done reports completed cancellation.
Shadow descriptor channel	Input/output	cfg_index: 6 bits; cfg_ wdata: 32 bits; controls: 1 bit	cfg_valid with cfg_commit=0 writes a shadow word; cfg_commit=1 validates without writing. cfg_ready requires quiescence and current derived validation. cfg_done/cfg_error report commit acceptance or rejection; rejection preserves the active image.
Draft verification channel	Input/output	draft_token0/ draft_token1/ draft_pos: 32 bits; verify_ accepted: 2 bits	draft_valid/draft_ready accept two tokens with draft_ new_sequence. verify_valid, verify_accepted and verify_error remain held until verify_ready. Accepted length is unsigned 0, 1 or 2. Legacy draft_ready is zero.
Banked memory requests	Output with input readiness	B valid/write/ ready bits; kind 4B; projection 3B; layer/row/col/ wdata 32B; tag 64B	Signals use mem_req_* names. Bank zero occupies the least-significant slice. Requests retain semantic coordinates and stable payloads until acceptance.

Signal / group	Direction	Width	Contract
Banked memory responses	Input with output readiness	B valid/error/ready bits; rdata 32B; tag 64B	Signals use mem_rsp_* names. Responses may reorder by tag, including within a bank. Writes also require acknowledgment. Unknown, duplicate or wrong-epoch tags fault and drain.
FP32 arithmetic leaf	Internal reusable ready/valid interface	op: 4; a/b/result: 32; flags: 5 bits	Includes clk, rst_n and flush. Opcodes 0–9 select ADD, SUB, MUL, DIV, SQRT, BF16 widening, FP16 widening, BF16 narrowing, FP16 narrowing and signed-int32 conversion. Other opcodes return canonical NaN with invalid.

Parameter	Default	Meaning / supported values
BANKS	16	Physical banks: 1, 2, 4, 8 or 16. Runtime active banks cannot exceed physical banks; legacy modes use at most eight.
QUEUE_DEPTH	3	1–16 ownership slots per bank; returned but unconsumed responses retain ownership.
TILE_K / TILE_ROWS	128 / 8	Input tile length 1–128; parallel output rows 1–8. Tail lanes are masked. Native STA used 8 / 2, not these defaults.
Runtime geometry	Qwen2.5-7B AWQ	Layers 1–64; hidden 2–8192; intermediate 1–32768; query/KV heads 1–128; even head dimension 2–256; vocabulary 1–262144; context 1–131072; group size 1–128. Require hidden=query_heads*head_dim and query_heads=ratio*KV_heads, ratio in {1,3,6,7,8}.
Reset profile	28 layers; H3584; I18944; Q28/KV4; D128; vocabulary152064	BF16, asymmetric W4A16, group128, QKV bias enabled, context32768, theta1000000, epsilon1e-6, page_tokens32 and 1024 physical pages. Active banks default to BANKS.
Format and paging selection	BF16 cache; A16	BF16/FP16 inputs; native/W4/W8 weights; A8 only with W4/W8. Q4 cache BF16 or INT8. Page tokens 1/2/4/8/16/32; physical pages 1–131072 with sufficient capacity and valid coordinates.

III Behavior and corner cases

A Descriptor ABI and switching

Words 0–23 encode ABI, mode, activation format, weight format, activation quantization, cache format, banks, layers, hidden, intermediate, query heads, KV heads, head dimension, vocabulary, context, group size, bias, theta, epsilon, page tokens, physical pages, zero convention, ratio and profile label. Word24 is zero in Q4 and legacy cache-block length otherwise; words25–31 are zero. Indices >=32 record a write fault. Successful activation invalidates sequence/prediction/cache epochs; rejected activation does not.

B IEEE arithmetic

FP32 arithmetic uses round-to-nearest ties-to-even, gradual underflow, signed zero, infinities and canonical NaN 0x7fc00000. Flags are invalid/divide-by-zero/overflow/underflow/inexact from bit4 to bit0. Multiplication and addition round separately, with increasing-index reductions, never fused FMA. Results appear after acceptance E+8 for ADD/SUB/MUL, E+2 for conversions and E+64 for DIV/SQRT; backpressure holds the single result slot.

C Transformer mathematics

RMSNorm computes gamma*x/sqrt(sum(x*x)/hidden+epsilon). RoPE pairs split halves and requests generation/position-qualified coefficients. Contiguous query-head groups share KV heads. Attention includes the current token, scales by 1/sqrt(head_dim) and uses max-subtracted softmax. SwiGLU computes down(silu(gate)*up). Exponential uses sequential range reduction and a degree-16 polynomial; SiLU uses exp(-abs(x)) to avoid avoidable overflow.

D Quantization boundaries

W4/W8 compute FP32(q-z)*scale, narrow once to BF16, then widen for multiplication. AWQ logical lanes use offsets [0,16,4,20,8,24,12,28]; asymmetric zeros encode z directly. W8 bytes ascend from the least-significant byte. A8 uses per-tile maxabs/127, zero scale1 and RNE codes clamped to [-127,127]. Groups continue across tile boundaries and restart per output row.

E Memory ABI

Q4 kinds 0−15 are embedding/lm-head, weights, zeros, scales, bias, attention gamma, FFN gamma, final gamma, scratch, BF16 K, BF16 V, RoPE, INT8 K, INT8 V, cache metadata and page table. Projections Q/K/V/O/gate/up/down are 0−6; kind0 projection0/1 selects embedding/lm-head. BF16/FP16 scalar tables occupy low16 bits; FP32 scratch/scales use all32. Q4 bank=(row*kind_stride+col)&(active_banks-1); SPEC.md defines each stride and the distinct legacy layout.

F Scratch and paged history

Scratch slots 0−12 hold residual, attention norm, Q, K, V, context, O, attention residual, FFN norm, gate, up, nonlinear product and down; slot13 is reserved and slot14 final norm. Page entries use layer=epoch,row=logical_page,col=0, with valid bit31. Cache row=physical_page*page_tokens+offset. BF16 columns are head*D+dimension; ordinary INT8 columns pack four ascending bytes. Per-head K/V metadata is published only after payload acknowledgment.

G Exceptional cache encoding

INT8 heads containing nonfinite values, negative zero or a nonzero maximum whose scale rounds to zero use FP32 escape payloads, selected by metadata 0x7fc00000. Escape projection1 preserves finite bits and signed infinities/zeros while canonicalizing NaNs. Ordinary -128 codes, malformed scales, padding, escape placeholders and noncanonical escape NaNs are rejected. This avoids silent exceptional-value loss.

H Speculation, reset and cancellation

Draft verification compares hardware target argmax predictions, with lowest-index tie breaking, and commits only the accepted prefix when its held result transfers. Failed or cancelled batches commit no new prefix; tentative new-sequence pages cannot alias protected committed pages. Cancellation suppresses new requests/ results and drains accepted transactions. A token accepted at E0 cannot transfer its first request before E2. Final-logit retirement produces one completion cycle before token readiness returns.

IV Verification methods and measured results

A Software-vs-Hardware numerical comparison

Method. Compiled RTL against independent rational IEEE primitives, 70-digit nonlinear mathematics and public intermediate/logit expectations.

Result. 24 system cases passed across seeds17/29/43, including all five ratios per seed, H64 cases, two compact 28-layer profile-family cases and singleton geometry. Stage tolerance was 2e-4+2e-4*abs(reference); logits 2e-3+2e-3*abs(reference), cosine >=0.9999. These are enforced bounds, not reported worst-case errors.

Evidence: E126, E093

B Primitive and quantization correctness

Method. Seeded exact comparisons, exhaustive 65536-encoding widening for each 16-bit format, directed exceptions, latency/stall/flush tests and nonlinear interval sweeps.

Result. 234088 arithmetic comparisons passed; 2057 exponential, 2057 SiLU, 7332 dequantization and 4686 A8 comparisons passed. Exponential/SiLU checks enforced the specified 2e-6-scale bounds; all five bank selections received tensor tests.

Evidence: E125

C Protocol, compatibility and model geometry

Method. Public-pin stalls/errors/cancellation, speculative lengths0/1/2, tagged scheduling, independent legacy integer models and authenticated configuration fixtures.

Result. System and compatibility suites passed. Legacy primary coverage comprised 21 cases across three seeds and supported cache modes. Full-shape tests exercised eight GEMV output rows plus boundary addresses, not complete full-model inference.

Evidence: E126, E127

D Five-domain benchmarks

Method. 17 seed17 mode/bank cases with independent correctness, cycle/traffic accounting and architecture-switching execution.

Result. Passed. One-bank compact Q4 measured four tokens, 68 logits, 996 intermediate comparisons, 468710 cycles, 11034 reads and 996 writes: 853.41 simulated tokens/s at100MHz. Domain spans include intervening work; representation-dependent cases do not establish cross-generation speedup.

Evidence: E128

E Defect sensitivity

Method. Disposable compiled mutations targeting ten numerical, mapping, descriptor, routing, cache and roll-back classes.

Result. All ten injected classes triggered specific executed independent mismatch checks. Compilation failures alone were not accepted as detection.

Evidence: E129

F Synthesis and portable reproduction

Method. Yosys hierarchy/process/memory lowering and structural assertions; fresh declared-source copy running Icarus token smoke, compiled end-to-end execution, frontend comparisons and synthesis.

Result. Passed. Default16-bank, one-bank and compatibility designs retained required functions with zero-latch/div/mod assertions. Generic cell counts were168785,159681 and41081 respectively; these are not physical area.

Evidence: E130, E131

G L1 RTL timing lint

Method. Native hazard analysis of42 delivered RTL files.

Result. Passed: no detected combinational divider/modulo, mixed sequential assignment, LUT-clock-gating or AST loop-unroll hazards.

Evidence: E133

H L2 Intel Quartus Prime17.1 TimeQuest STA

Method. Native100MHz,10.0ns analysis; post_map netlist, plan_plus_post_map_sta, TILE_K8/TILE_ROWS2.

Result. Passed: WNS_setup = +2.925 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 141.34 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E134

I L3 physical Arria10 FPGA HIL

Method. Three consecutive physical PCIe bridge register57 probes.

Result. Gate passed3/3 deterministically. Receipt tier is L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; it does not establish accelerator bitstream deployment, workload execution or board numerical comparison.

Evidence: E135

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	9 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 50	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

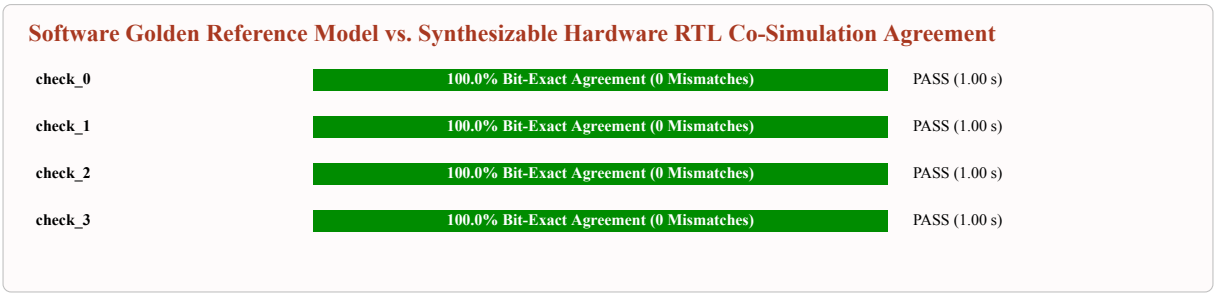


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +2.925 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

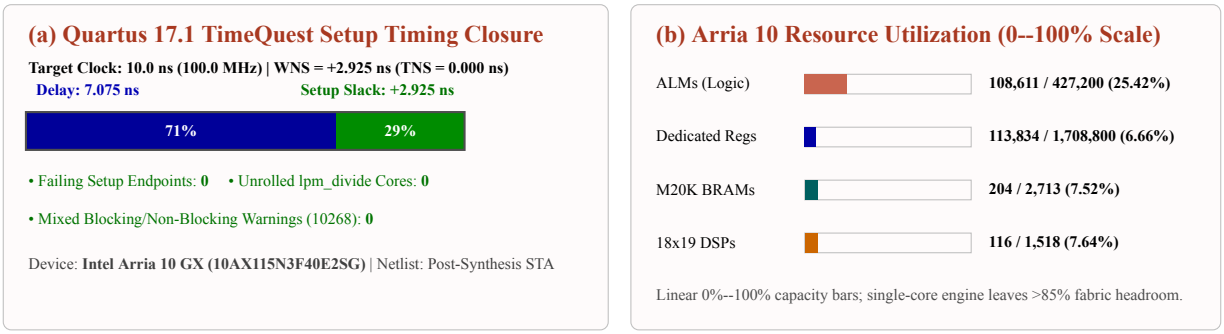


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

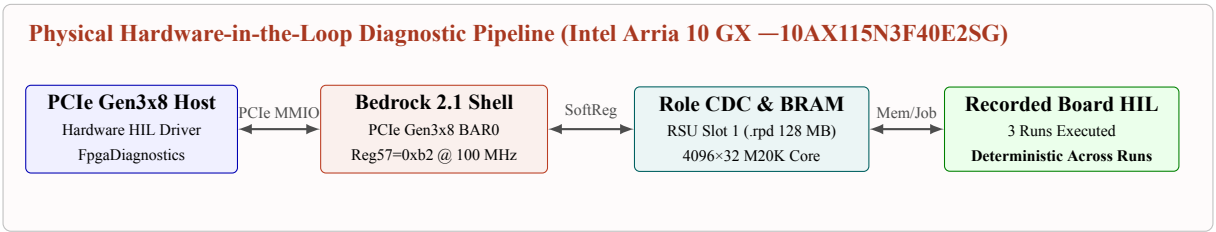


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/, design/ and tests/ with Python3, Icarus/vvp, Verilator, a C++17 compiler, make and Yosys on PATH. Compile anvil_q3_00_pkg.sv and anvil_q4_00_fp_pkg.sv before dependent modules; select anvil_q4_top. The parent reference directory is not a delivered-build dependency.

Run python3 tests/verify.py for all L1 suites, or select smoke, arithmetic, system, compatibility, benchmarks, mutations, synthesis or portable with --suite. tests/README.md documents simulator tiering and fresh receipt generation. Portable reproduction actually executed the bounded subset described above, not the entire matrix.

Supply semantic table storage, generation/position-correct RoPE coefficients, scratch, valid nonaliased page mappings and tagged acknowledgments for writes as well as reads. Keep mappings fixed while active and reset bridge transaction epochs with the DUT.

After reset, use the Q4 reset profile or write a complete shadow descriptor and commit while cfg_ready is asserted. Wait for cfg_done and reject cfg_error before submitting a sequence. Explicitly activate legacy descriptors rather than assuming legacy reset defaults.

The retained system timeout was repaired through bounded bridge storage, simulator scheduling/compiler improvements and case concurrency without changing RTL, numerical models or thresholds. Later admitted suites passed; failed evidence remains retained. Cache payload-before-metadata acknowledgment and exceptional-head escape address distinct data-publication and numerical-preservation risks.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 42 Verilog/SystemVerilog files; 6797 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/anvil_q3_00_pkg.sv	100 total	93 code
rtl/anvil_q3_accumulator.sv	66 total	64 code
rtl/anvil_q3_add128.sv	45 total	45 code
rtl/anvil_q3_awq_dot8.sv	95 total	91 code
rtl/anvil_q3_cache_codec.sv	252 total	245 code
rtl/anvil_q3_divider.sv	114 total	112 code
rtl/anvil_q3_elementwise.sv	122 total	121 code
rtl/anvil_q3_exp.sv	131 total	128 code
rtl/anvil_q3_gqa_attention.sv	235 total	234 code
rtl/anvil_q3_isqrt.sv	84 total	82 code
rtl/anvil_q3_kv_cache.sv	246 total	242 code
rtl/anvil_q3_mem_port.sv	77 total	75 code
rtl/anvil_q3_multiplier.sv	99 total	97 code
rtl/anvil_q3_rmsnorm.sv	193 total	190 code
rtl/anvil_q3_rope.sv	144 total	142 code
rtl/anvil_q3_signed_divider.sv	81 total	80 code
rtl/anvil_q3_silu.sv	118 total	115 code
rtl/anvil_q3_token_io.sv	125 total	124 code
rtl/anvil_q4_00_fp_pkg.sv	307 total	289 code
rtl/anvil_q4_activation_quantizer.sv	86 total	86 code
rtl/anvil_q4_bank_scheduler.sv	414 total	398 code
rtl/anvil_q4_compat.sv	201 total	200 code
rtl/anvil_q4_decoder_layer.sv	217 total	216 code
rtl/anvil_q4_dequant.sv	165 total	158 code
rtl/anvil_q4_descriptor.sv	266 total	252 code
rtl/anvil_q4_elementwise.sv	104 total	104 code
rtl/anvil_q4_exp.sv	133 total	131 code
rtl/anvil_q4_fp32.sv	330 total	322 code
rtl/anvil_q4_gemv.sv	270 total	270 code
rtl/anvil_q4_gqa_attention.sv	233 total	232 code
rtl/anvil_q4_kv_cache.sv	331 total	331 code
rtl/anvil_q4_legacy_gemv.sv	151 total	151 code
rtl/anvil_q4_mem_port.sv	74 total	74 code
rtl/anvil_q4_rmsnorm.sv	138 total	138 code
rtl/anvil_q4_rope.sv	127 total	127 code
rtl/anvil_q4_silu.sv	86 total	86 code
rtl/anvil_q4_speculative.sv	94 total	94 code

Production file	Physical lines	Code lines
rtl/anvil_q4_swiglu_mlp.sv	117 total	117 code
rtl/anvil_q4_tensor_array.sv	194 total	194 code
rtl/anvil_q4_token_io.sv	139 total	139 code
rtl/anvil_q4_top.sv	360 total	357 code
rtl/anvil_q4_umul64.sv	51 total	51 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E003	design/SPEC.md
E040	rtl/anvil_q4_speculative.sv
E043	rtl/anvil_q4_token_io.sv
E044	rtl/anvil_q4_top.sv
E093	tests/transformer_reference.py
E125	.vulcan/engineering/run-0001/attempt-0036/stdout.log
E126	.vulcan/engineering/run-0001/attempt-0037/stdout.log
E127	.vulcan/engineering/run-0001/attempt-0038/stdout.log
E128	.vulcan/engineering/run-0001/attempt-0039/stdout.log
E129	.vulcan/engineering/run-0001/attempt-0040/stdout.log
E130	.vulcan/engineering/run-0001/attempt-0041/stdout.log
E131	.vulcan/engineering/run-0001/attempt-0042/stdout.log
E133	.vulcan/engineering/run-0001/rtl-timing-lint.json
E134	.vulcan/engineering/run-0001/quartus-sta.json
E135	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).